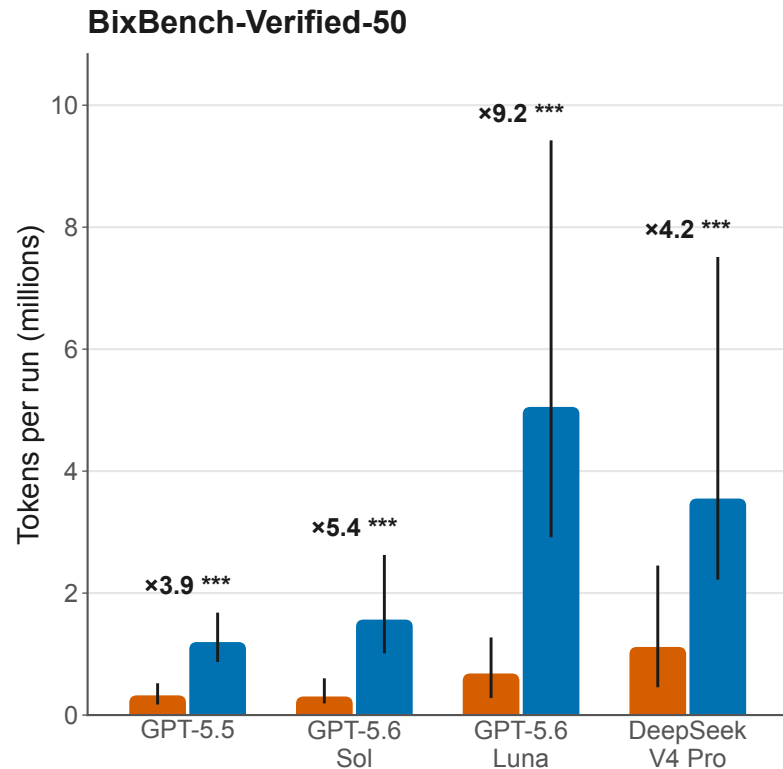


Galaxy runs use 3–9× the tokens of custom code on BixBench and CompBioBench

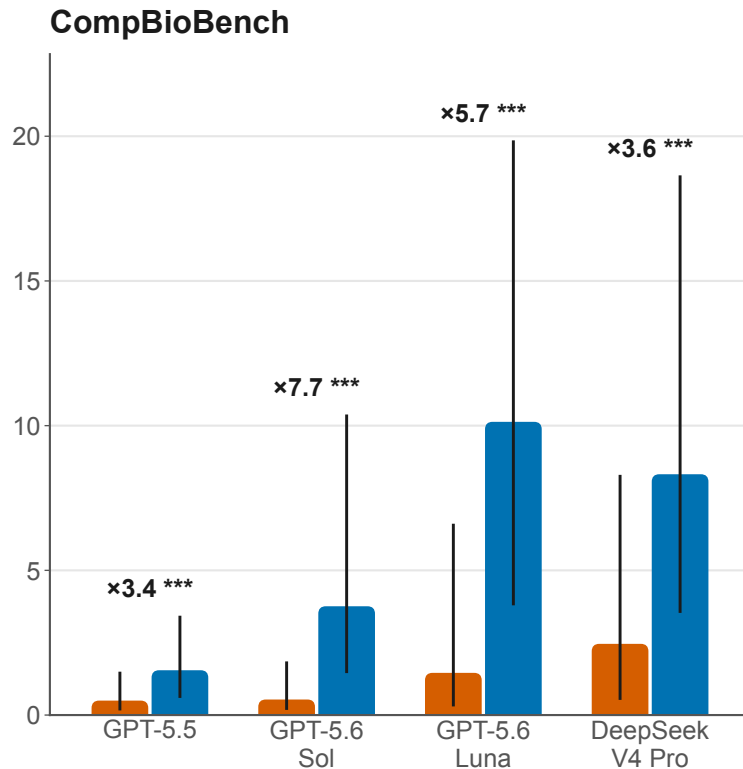
Bars: median tokens per run (input including cached context, plus output); whiskers: interquartile range. Above each pair: Galaxy ÷ custom code, the geometric mean over tasks of the ratio of their mean tokens (replicates averaged), with the paired cluster sign-flip test, Holm-adjusted over the 12 model comparisons:

* adj. $P < 0.05$, ** < 0.01 , *** < 0.001 , ns not significant. Under each panel: the same ratio for the four models together, and the share of input tokens read from the cache.

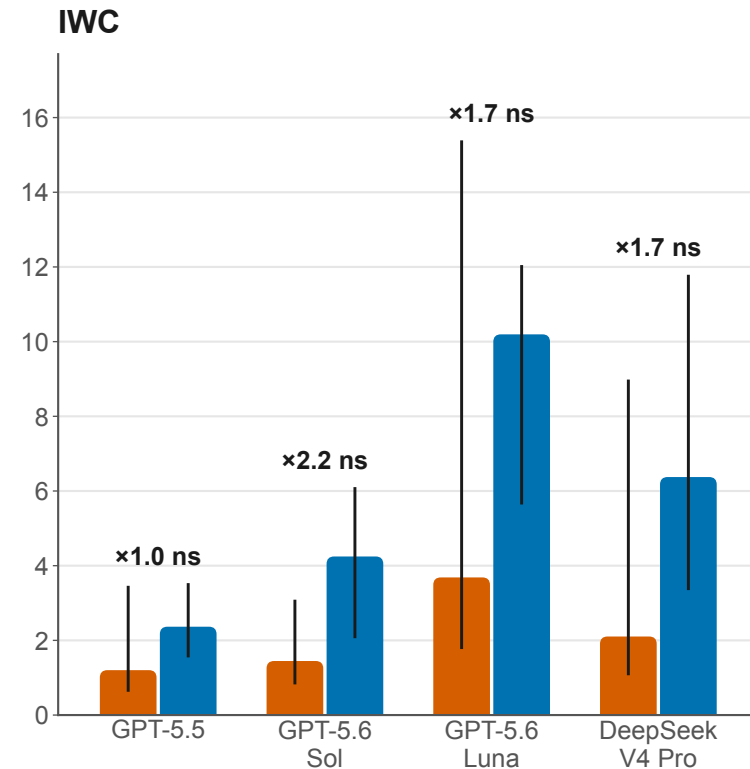
Custom code Galaxy



All four models: ×5.3 (95% CI 4.0 to 7.1), $P < 0.001$
Cached share of input: custom code 96%, Galaxy 98%



All four models: ×4.8 (95% CI 4.0 to 5.8), $P < 0.001$
Cached share of input: custom code 97%, Galaxy 97%



All four models: ×1.6 (95% CI 0.8 to 3.1), $P = 0.26$
Cached share of input: custom code 98%, Galaxy 93%